Revision Questions: Cell Structure and Organisation

Biology · Cell Structure and Organisation · Section A: multiple choice · Section B: short answer and free response

Name: _____ Class: _____ Date: _____

Test yourself on Topic 1 (lessons 1.1 and 1.2). Section A has 10 multiple-choice questions. Section B has short and longer written answers.

Section A: Multiple choice (10 questions)

Choose the best answer. You can check each answer and try again.

1. Which structure is partially permeable?

- ○ Cell membrane
- ○ Chloroplast
- ○ Cell wall
- ○ Cell sap

2. Which structure carries hereditary information?

- ○ Mitochondrion
- ○ DNA in the nucleus
- ○ Cytoplasm
- ○ Ribosome

3. Which pair of organelles can only be seen clearly with an electron microscope?

- ○ Cell membrane and cytoplasm
- ○ Nucleus and vacuole
- ○ Mitochondria and ribosomes
- ○ Chloroplasts and cell wall

4. What does cell sap contain?

- ○ DNA
- ○ Chlorophyll
- ○ Cellulose
- ○ Water with dissolved sugar, mineral salts and amino acids

5. Which stain is used to see the nucleus in human cheek cells?

- ○ Methylene blue
- ○ Cell sap
- ○ Chlorophyll
- ○ Iodine solution

6. A cell has a nucleus, a cell membrane, many small vacuoles and no cell wall. What type of cell is it?

- ○ An onion cell
- ○ An animal cell
- ○ A root hair cell
- ○ A leaf cell

7. Which structure is the site of photosynthesis?

- ○ Mitochondrion
- ○ Ribosome
- ○ Chloroplast
- ○ Nucleus

8. Which feature of a muscle cell supplies more energy for contraction?

- ○ A cell wall
- ○ A long extension
- ○ No nucleus
- ○ Many mitochondria

9. Which feature of a root hair cell increases the rate of absorption of water and mineral salts?

- ○ A long, narrow extension giving a large surface area
- ○ Many chloroplasts
- ○ A thick cell wall
- ○ No nucleus

10. Why can a red blood cell transport more oxygen than if it had a nucleus?

- ○ It has more mitochondria.
- ○ It has more space for haemoglobin.
- ○ It has chloroplasts.
- ○ It has a cell wall.

Section B: Short answer, structured and free response (7 questions)

Write your answer in the box and check it against the mark scheme. Use full sentences and show your working in calculations, with units. The mark scheme shows which points earn marks.

1. State the function of (a) the nucleus and (b) the ribosomes. [2 marks]

Write your answer here. Do not type your name or any personal details.

2. Compare the structure of a typical plant cell with a typical animal cell. Give one similarity and two differences. [3 marks]

Write your answer here. Do not type your name or any personal details.

3. A student says, "The cell wall controls what enters and leaves a plant cell." Explain why the student is wrong. [2 marks]

Write your answer here. Do not type your name or any personal details.

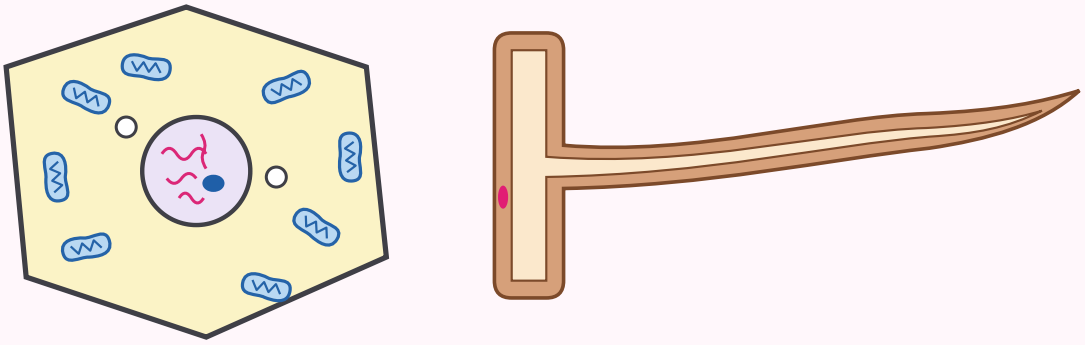
4. A red blood cell transports oxygen. Explain how two features of its structure adapt it to this function. [3 marks]

Write your answer here. Do not type your name or any personal details.

5. Explain how a root hair cell and a muscle cell are each adapted to their functions. [3 marks]

Write your answer here. Do not type your name or any personal details.

6. Fig. 6.1 shows a liver cell (A) and a root hair cell (B). They are not drawn to scale.



A B

Fig. 6.1

6(a). Which cell, A or B, is a plant cell? Give one reason from Fig. 6.1. [2 marks]

Write your answer here. Do not type your name or any personal details.

6(b). State two differences in structure between the liver cell and the root hair cell.
[2 marks]

Write your answer here. Do not type your name or any personal details.

6(c). Liver cells contain many mitochondria. Suggest why. [2 marks]

Write your answer here. Do not type your name or any personal details.

6(d). Explain how the root hair cell in Fig. 6.1 is adapted to absorb water from the soil. [2 marks]

Write your answer here. Do not type your name or any personal details.

7. A typical plant cell, such as a leaf cell, has chloroplasts. Explain why a root hair cell does not have chloroplasts. [2 marks]

Write your answer here. Do not type your name or any personal details.